Adversarial Exclusive Completions for a Common-Ambient Prime-Shell Holdout

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Abstract

The common-ambient union-shell holdout of TPC-307 exposed three finite discordances between a constrained source-budget orientation and the error on withheld exclusive shell pieces. We stress that observation with an explicit adversarial completion envelope: for each fixed directional prediction, enumerate every binary relabelling within Hamming radii $r = 0, 1, 2$ on the corresponding exclusive holdout. The Hamming-ball construction, its binomial cardinality, fixed-prediction extrema, radius monotonicity, radius-zero recovery, and sign invariance are exact finite facts. On the locked 18-cell twin-prime diagnostic spine, the producer and an independent NumPy replay reproduce 54 envelope observations and 702 candidate evaluations. The agreement census changes from 13/3/2 to 11/2/5 to 10/1/7 (concordant/discordant/unresolved) as $r = 0, 1, 2$; every surviving discordance remains on $Q = 70 \to 90$ with kernel exponent one. The envelope therefore attenuates but does not remove the finite obstruction, while widening several cells into the unresolved band. The physical computation is a numerical replay with padded decimal enclosures, not a directed-rounding certificate. No causal, asymptotic, arithmetic, or twin-prime conclusion is claimed.

1 Question and frozen parent object

TPC-307 placed each adjacent prime-shell pair in one common union ambient operator, fitted two aligned directional targets on the overlap, and used the exclusive pieces as withheld holdouts [1]. Its finite atlas contained three budget/holdout discordances, all on the final transition $70 \to 90$ at exponent one. A natural concern is that the native exclusive labels might be a particularly favorable or unfavorable completion. The minimal next question is therefore adversarial and finite:

> Does the localized discordance survive when each native exclusive label is replaced by every nearby binary completion, while the fitted prediction and source budget remain fixed?

We retain the parent conventions without adding a moving parameter. The source interval is $[257, 512] \cap \mathbb{Z}$, the height is $H = 58$, and the source comparison cutoff is $z = 5$. The shell spine is $Q = (50, 60, 70, 90)$, with adjacent pairs $(50, 60), (60, 70), (70, 90)$. The two literal kernel exponents are $e = 1, 2$, and the relative overlap-fit tolerances are $\tau \in \{0.25, 0.5, 0.75\}$. The profile matrix has the 17 cutoffs

$$3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61.$$

For an adjacent pair, write U for the union shell, O for the overlap, and E_L, E_R for the left and right exclusive pieces. TPC-307 fixes the common ambient matrix V_U, the overlap-only frontier coefficients c_L, c_R, and the common prefix $k = \max(k_L, k_R)$. It also fixes the aligned native binary targets $h_L \in \{-1, +1\}^{|E_L|}$ and $h_R \in \{-1, +1\}^{|E_R|}$. No TPC-308 completion is fed back into the coefficient fit or into the budget comparison. The source-first labels come from the physical-Gram-dependent construction inherited from TPC-302 [2]; that leakage is retained in the claim firewall.

2 Completion envelopes

For a binary vector h of length m and a nonnegative integer r, define the finite Hamming ball

$$\mathcal{C}_r(h) = \{h' \in \{-1,+1\}^m : d_H(h',h) \le r\}.$$

If y is the already fitted prediction on the same exclusive rows, define the lower and upper completion losses

$$L_r^-(y,h) = \min_{h'\in\mathcal{C}_r(h)} \frac{1}{m}\|y-h'\|_2^2, \qquad L_r^+(y,h) = \max_{h'\in\mathcal{C}_r(h)} \frac{1}{m}\|y-h'\|_2^2. \tag{1}$$

The producer enumerates a flip subset F and changes h_j to $-h_j$ for $j \in F$. Thus the candidate count is

$$|\mathcal{C}_r(h)| = \sum_{j=0}^{\min(r,m)} \binom{m}{j}. \tag{2}$$

Proposition 1 (Finite envelope algebra)**.** *For fixed y and binary h, the flip-subset enumeration gives the exact minimum and maximum in (1). Further,*

$$L_{r+1}^-(y,h) \le L_r^-(y,h) \le L_r^+(y,h) \le L_{r+1}^+(y,h), \tag{3}$$

and $\mathcal{C}_0(h) = \{h\}$. Simultaneously replacing (y,h) by $(-y,-h)$ preserves every completion loss.

Proof. For any completion h', let $F = \{j : h'_j \ne h_j\}$. Since the coordinates are binary, $h'_j = -h_j$ exactly on F, and the Hamming constraint is $|F| \le r$. Conversely, every such F produces one completion, and the recovered difference set is unique. This proves the bijection and (2), so the finite list contains all and only feasible points and its extrema are exact.

The inclusion $\mathcal{C}_r(h) \subseteq \mathcal{C}_{r+1}(h)$ gives (3) by the elementary monotonicity of a minimum and a maximum under set inclusion. The only subset of size at most zero is the empty set, proving the singleton claim. Finally, $\|(-y)-(-h')\|_2^2 = \|y-h'\|_2^2$, and global negation bijects the two completion balls. ☐

For positive losses, TPC-308 records the conservative right-over-left interval

$$R_r \in \left[\frac{L_r^-(y_R,h_R)}{L_r^+(y_L,h_L)}, \frac{L_r^+(y_R,h_R)}{L_r^-(y_L,h_L)}\right]. \tag{4}$$

The lower endpoint is not interpreted as a point estimate. An interval wholly below 0.9 is called right-lower, one wholly above 1.1 is left-lower, and all other intervals are unresolved.

Proposition 2 (Conditional class soundness)**.** *If the four losses in (4) are positive and the displayed endpoints are valid, then an upper endpoint below* 0.9 *implies a strict right-lower classification, while a lower endpoint above* 1.1 *implies a strict left-lower classification.*

Proof. For positive $a \in [a_-,a_+]$ and $b \in [b_-,b_+]$, $a/b \in [a_-/b_+, a_+/b_-]$. Apply this to the right and left finite loss intervals and then use the definitions of the two strict classes. ☐

Remark 1. *The completion ball is a set of adversarial diagnostics, not a distribution and not a model for how arithmetic labels are generated. In particular, (1) does not create causal separation between the physical operator and the target.*

Table 1: Agreement census over all 18 cells.

radius r	envelope observations	candidates	concordant	discordant	unresolved
0	18	36	13	3	2
1	18	186	11	2	5
2	18	480	10	1	7

Table 2: Discordances by transition and completion radius.

transition	$r = 0$	$r = 1$	$r = 2$
$50 \to 60$	0	0	0
$60 \to 70$	0	0	0
$70 \to 90$	3	2	1

3 Numerical protocol

For each of the 18 parent cells, the producer loads the locked TPC-307 certificate, reconstructs the six union shells, and reuses the two fixed predictions. It applies (1) independently to E_L and E_R at each of the three radii. The exclusive cardinality pairs are $(2, 5)$ on $50 \to 60$, $(2, 4)$ on $60 \to 70$, and $(5, 7)$ on $70 \to 90$; each transition occurs for six (e, τ) cells. Equation (2) consequently predicts the aggregate candidate totals 36, 186, and 480 at radii 0, 1, and 2.

The parent physical rows are assembled by the literal deleted-diagonal formula in vectorized float64 arithmetic. The parent frontier is then replayed at high precision on decimalized entries, and each reported value is padded by relative radius 10^{-5}. The independent checker does not import the TPC-308 producer: it rebuilds the source profiles and physical rows from the frozen TPC-268 engine, solves the overlap frontier with a separate NumPy implementation, enumerates the flip subsets, and checks the published enclosures with a 2×10^{-3} relative replay slack.

This separation supports numerical reproduction of a declared finite object. It does not support a formal interval certificate: neither the physical matrix construction nor the optimization path is evaluated with directed rounding. The exact statements are therefore limited to the finite protocol and its algebraic lemmas.

4 Results

Table 1 gives the central aggregate result. The first row recovers TPC-307 exactly at the class level, as required by $\mathcal{C}_0(h) = \{h\}$. Enlarging the completion ball reduces strict discordances from three to two to one, but also turns more cells unresolved.

The transition localization is shown in Table 2. No discordance occurs on either of the first two transitions at any tested radius. Every discordance is on $70 \to 90$ and exponent one; the count on that transition is $3, 2, 1$ as the radius increases.

At radius zero, the final transition's exponent-one ratios for the three tolerances are approximately 0.280, 0.511, and 1.156 for the holdout diagnostic, while the corresponding budget classes are respectively left, left, and right. These are the three parent discordances. At radius one, the first two remain strict right-lower holdout classes and the third becomes unresolved. At radius two, only the first remains strict right-lower; the other two are unresolved. Thus the adversarial envelope does not support a claim that the native discordance is wholly an artifact, but it does show that the strength of the class depends materially on the allowed completion radius.

The same finite replay produces seven unresolved cells at radius two. This is not a defect in the decision rule: broad extrema genuinely straddle the threshold band. It is evidence that an interval-aware obstruction map is more informative here than forcing a binary preference for every completion.

5 Interpretation and route status

The strongest positive result is an exact, auditable finite stress protocol: the fit geometry and source budget are frozen, the only changed object is a specified finite completion set, and the extrema and candidate counts have short proofs. The strongest obstruction is the surviving final-transition discordance at radius two. The strongest caution is the simultaneous growth of the unresolved region, which prevents a universal completion preference claim.

Several boundaries are important.

- The parent labels inherit a physical-Gram-dependent source-first construction from TPC-302; target-generation leakage is not removed.
- A finite Hamming envelope is not a probability law, a causal intervention, or an asymptotic family of shell completions.
- The padded decimal intervals are numerical replay enclosures, not a directed-rounding proof.
- No arithmetic L^2 estimate, fixed-power credit, uniform growing budget, full Route-B Gate B, or twin-prime conclusion follows.

The Session-named `propose.md` and Route-A/Route-B evaluator files are absent from this checkout. Accordingly, no official evaluator pass is asserted; the local fail-closed evaluation is recorded in the project route note and Bridge-B checker. The next minimal research question is to perturb the finite profile prefix around the selected k on the surviving cells. That tests a different stability axis without pretending that the current completion family identifies an arithmetic mechanism.

Reproducibility

The canonical certificate is `results/tpc308_certificate.json`. The producer, independent replay, exact rational stress suite, theorem ledger, claim firewall, and Bridge-B checker are in the project directory. Reproduction commands are listed in the project README. All empirical statements in this paper are finite observations under the locked protocol and should not be extrapolated to growing N or Q.

References

[1] Liang Wang. A common-ambient union-shell holdout for transported prime-shell labels, 2026. Local TPC-307 release in the prime-dynamics research repository.

[2] Liang Wang. Source-first growing-shell budget gap audit, 2026. Local TPC-302 release in the prime-dynamics research repository.